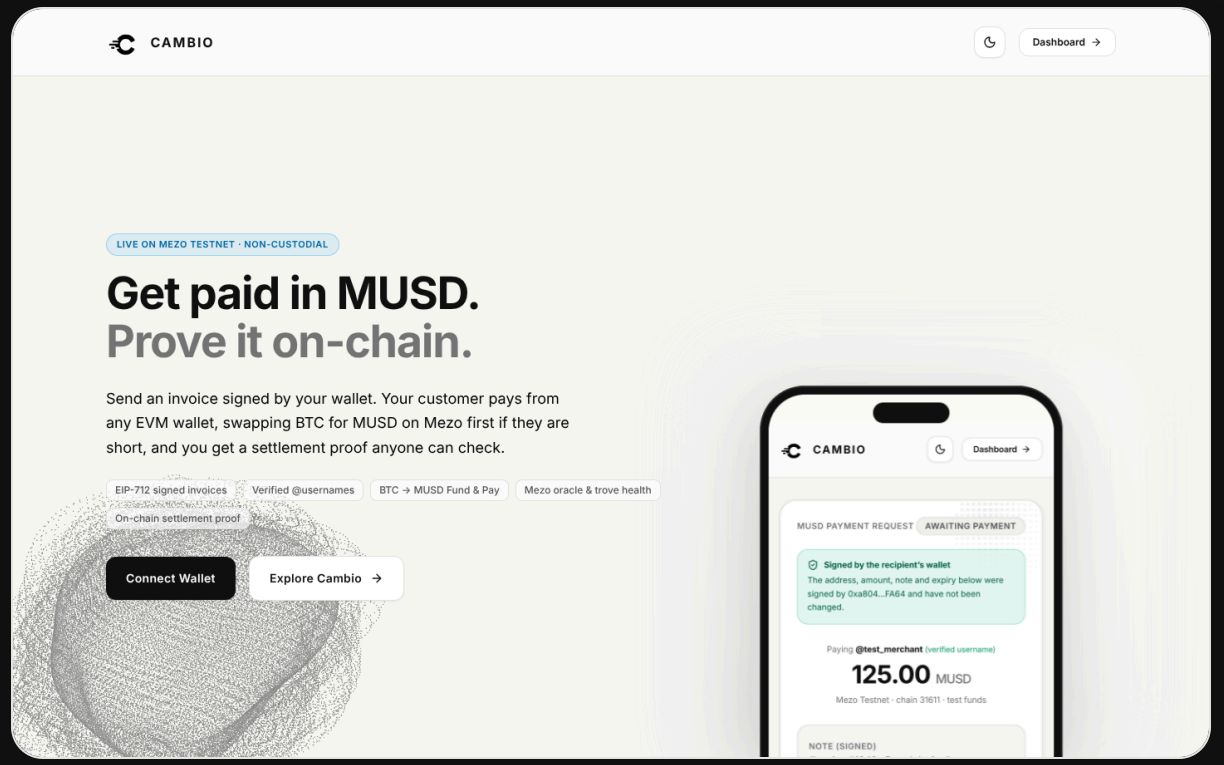
Cambio

Get paid in MUSD. Prove it on-chain.

Wallet-signed MUSD invoices any customer can pay, even with only BTC, and settlement that anyone can verify on Mezo.

Lushegs · solo developer

github.com/Lushegs01/cambio · cambio-five.vercel.app · Mezo Testnet (31611)



THE PAYMENT PROBLEM

Crypto invoicing still runs on copy-paste and screenshots

Links can be edited

A pasted address or a payment link can be changed in transit. The payer has no way to know the merchant wrote it.

"I paid you" is a screenshot

Merchants check explorers by hand. A payment is easy to miss, mis-attribute or double-count.

Payers hold BTC, invoices are in dollars

A BTC holder who owes a dollar amount must leave the invoice, find a DEX, size a swap and come back.

MUSD is Mezo's Bitcoin-backed dollar. Cambio makes it work as an **invoice currency** for people who already hold BTC.

WHO IT'S FOR

Start narrow: people who already get paid in crypto

Primary: crypto-paid freelancers and small studios

They invoice international clients and want a link nobody can edit, a dollar-stable asset, proof for bookkeeping, and no custody risk.

Secondary: Mezo-ecosystem merchants

Projects, communities and events whose customers already hold MUSD or BTC on Mezo.

Payers: BTC holders on Mezo

They pay a dollar invoice without leaving it, using Fund & Pay.

Persona: Ada, product designer

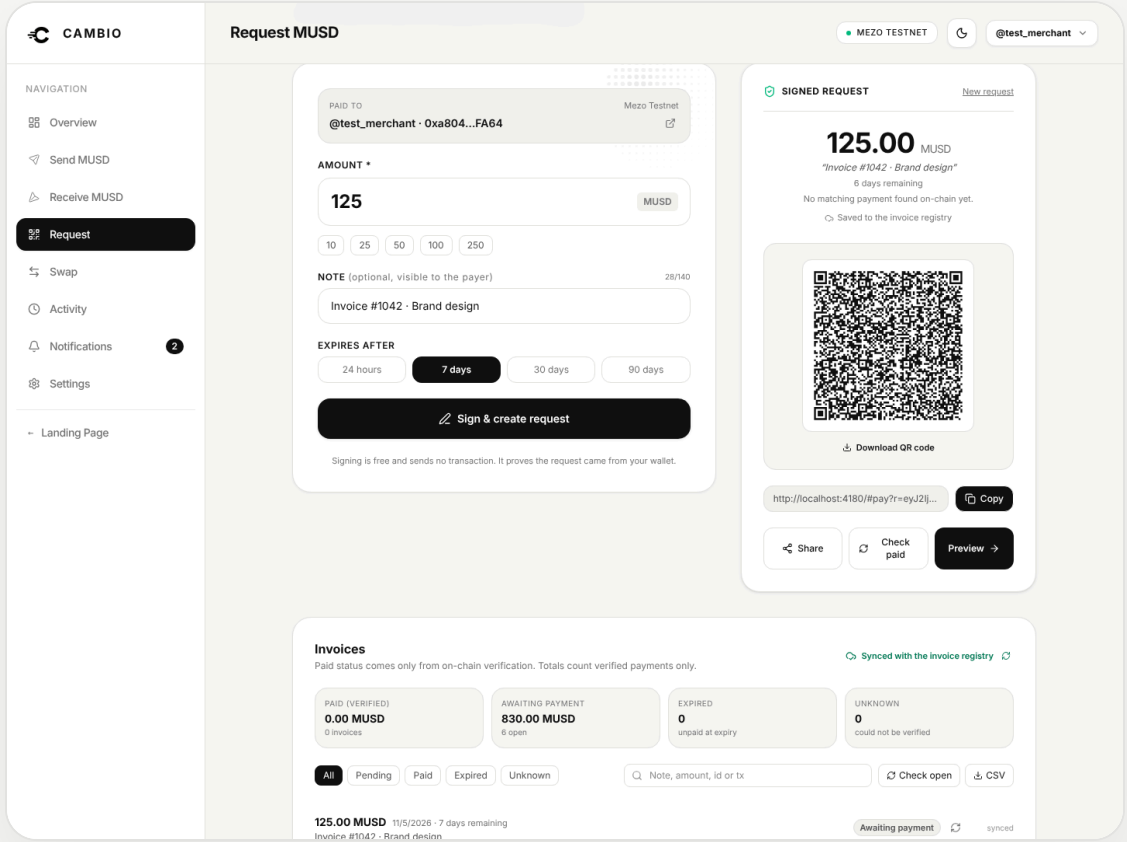
- 3–8 international clients a month, $300–$3,000 per invoice
- Today: pastes an address, asks for a screenshot, checks the explorer
- Needs: a tamper-proof link, a trusted "paid" status, a monthly export

These are hypotheses. User research has not started. We make no claim of mass-market or regional demand, and Cambio has no fiat off-ramp.

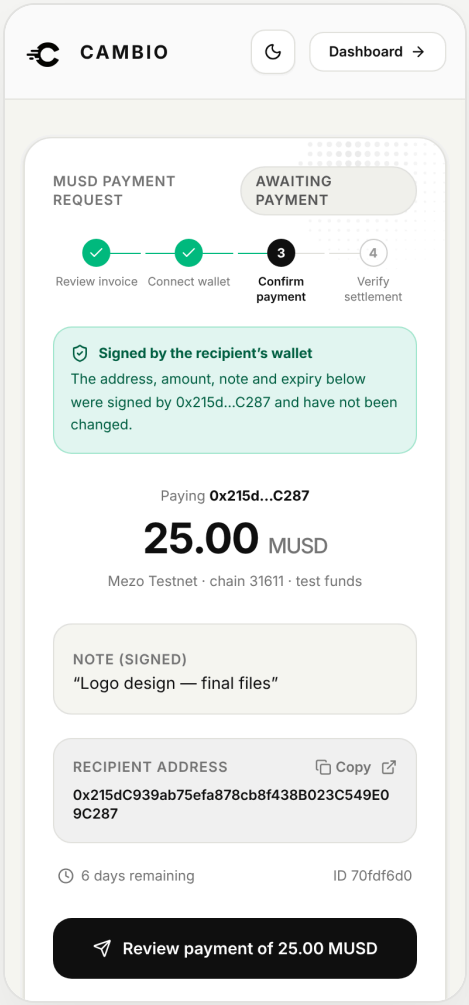
THE SOLUTION

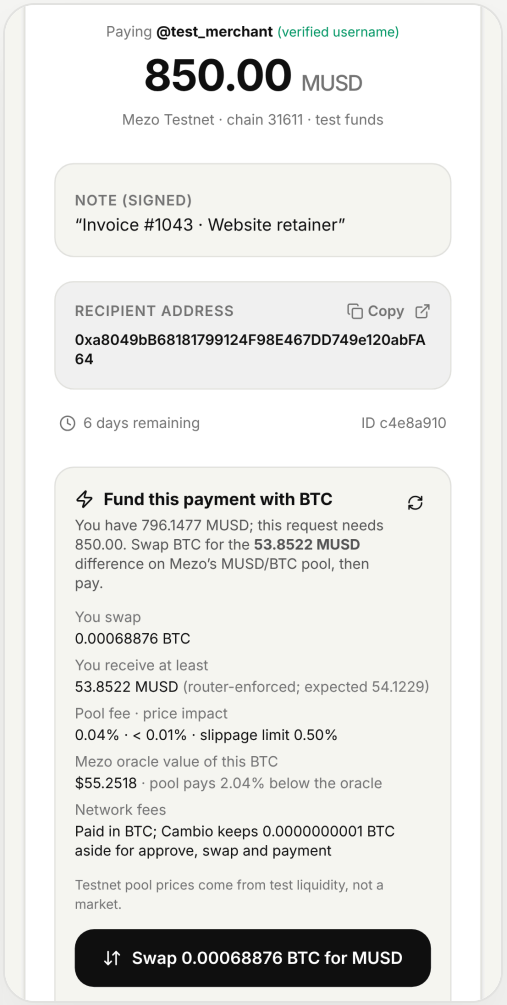
Signed invoices · Fund & Pay · verified settlement · merchant workspace

- **EIP-712 signed invoices.** The recipient, amount, token, chain, ID and expiry are signed by the merchant's wallet; the payer verifies the signature before Pay appears.
- **Verified @usernames.** Each name is bound by a server-verified SIWE signature.
- **Fund & Pay.** If the payer is short, Cambio sizes a BTC→MUSD swap for exactly the shortfall on the same page.
- **Public proof page.** The payment carries the invoice ID in its calldata; anyone opens a proof link and the browser re-runs every check, no wallet needed.
- **Invoice registry.** Syncs across devices; "paid" is only ever set by on-chain verification. Late/duplicate views, CSV.
- **Share anywhere.** WhatsApp, Telegram, email, QR, embeddable pay button — always the exact signed link.



Invoice → fund with BTC → pay → verified





1. The merchant signs an invoice. It's free, and the link is saved to the registry.
2. The customer opens it: a stepper guides them; the signature is verified and Cambio checks the invoice is not already paid.
3. They're 53.85 MUSD short, so Cambio quotes 0.00068876 BTC in, with at least 53.85 MUSD out (router-enforced). The Mezo oracle value is shown alongside.
4. Exact approval → swap → pay. Each step is its own wallet prompt.
5. The server re-reads the chain and marks the invoice **Paid · verified**; both sides get a public proof link. A refresh mid-payment resumes it, and Pay stays locked so nobody pays twice.

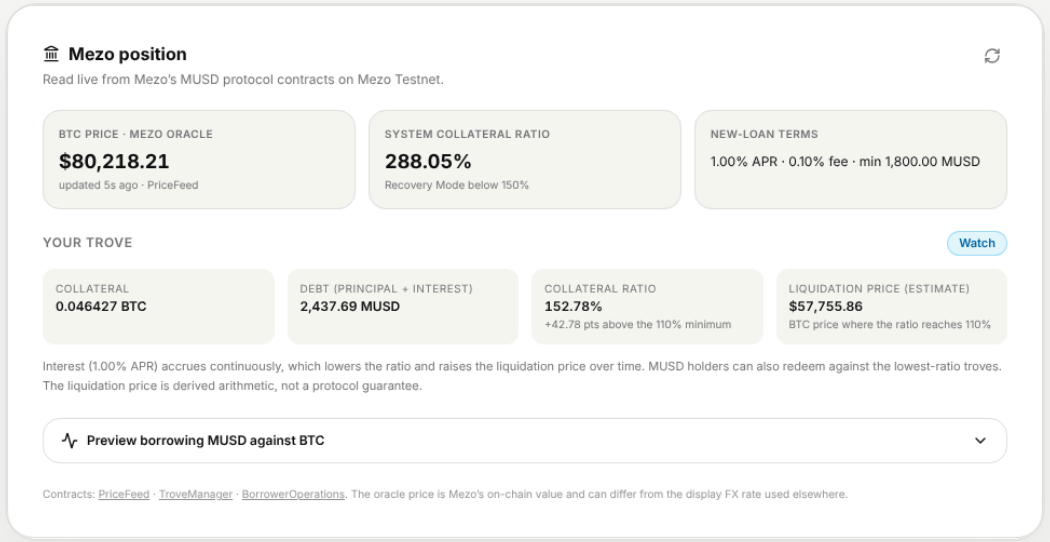
Left: production build, live testnet reads, test wallet (Nov 8). The Fund & Pay screen shows the real @lushegs wallet balance (796.1477 MUSD) against an 850 MUSD test invoice. Read-only capture; nothing was broadcast.

MEZO-NATIVE INTEGRATION

Built on Mezo's contracts, not a generic ERC-20

MEZO CONTRACT	CAMBIO USES IT FOR
MUSD	Transfers with a calldata-bound invoice ID; `Transfer`-event proof
Router · PoolFactory · MUSD/BTC pool	Quotes, on-chain fees, reserve-exact Fund & Pay sizing, exact-approval swaps
PriceFeed	BTC oracle price; a stale revert shows as stale, never as an off-chain substitute
TroveManager	Trove status, `getCurrentICR`, MCR/CCR, TCR, Recovery Mode
BorrowerOperations	Read-only `openTrove` preview: fee, minimum debt, gas compensation
Multicall3	17 → 5 RPC requests per refresh

Addresses come from the official `mezo-org/musd` deployments and are re-verified by `npm run verify:chain`. Cambio's ICR math equals a live `getCurrentICR` to the wei.

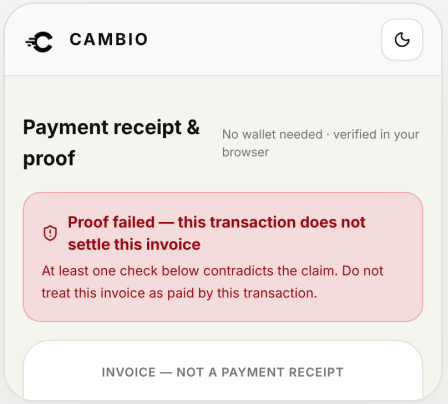


ON-CHAIN EVIDENCE (MEZO TESTNET)

Re-verified from the RPC with one command

PROOF	TRANSACTION	WHAT WAS CHECKED	VERDICT
MUSD transfer 10 MUSD, @solo → @lushegs	`0x8e39d309…5065e` `block 16390715`	Targets MUSD; `transfer` calldata; success; matching `Transfer` event; sender, recipient and amount	PASS
BTC → MUSD swap 0.01 BTC → 786.1477 MUSD	`0x4af2b0f0…33f6e` `block 16382064`	Mezo Router; BTC→MUSD route via PoolFactory; BTC into the pool; MUSD out ≥ router minimum (782.2170)	PASS
Signed-invoice settlement	`not yet recorded`	No transaction with Cambio's `CMB2` invoice-ID calldata exists yet (re-scanned Nov 8). E2E proves the wallet is asked to sign exactly that calldata; `--discover` records it automatically once paid.	PENDING

`npm run verify:evidence` → PASS / FAIL / INCONCLUSIVE; unreadable data is never a pass. The same checks run in the browser on every proof page. **Forgery test:** claiming the real transfer above as payment of a different signed invoice is rejected — it is a genuine MUSD transfer, but to the wrong recipient and without the invoice ID.



BUSINESS MODEL (HYPOTHESES TO VALIDATE)

Free to get paid. Merchant Pro for running a business on it.

Free: everything that exists today

Usernames, signed invoices, sharing, Fund & Pay, proof pages, merchant dashboard with sync and CSV

Merchant Pro: 9 MUSD / 30 days

price = hypothesis

Implemented on testnet. Unlocks a server-verified reconciliation report. Reminders and branding: to build.

Volume tier (later)

0.25% of verified volume above $2k/month (to test)

How Cambio actually collects (built)

The server issues an order (price, owner treasury, random id). The merchant's own wallet pays `transfer(treasury, price) || CMB2 || orderId`; the server verifies it on-chain and extends Pro atomically, once. No keys, no contracts, **no fee on payments**. Testnet: proves the mechanism, not revenue — no subscription paid yet.

Unit economics (estimates)

About $10–30/month fixed at pilot scale, about $115–265/month at scale. Roughly 4 Pro merchants cover the pilot and 15–30 cover scale. Marginal cost per invoice is under a cent.

TRACTION & VALIDATION

Honest status: zero users so far. The pilot is ready.

0

paying customers, revenue, or completed interviews. We claim none.

2 / 3

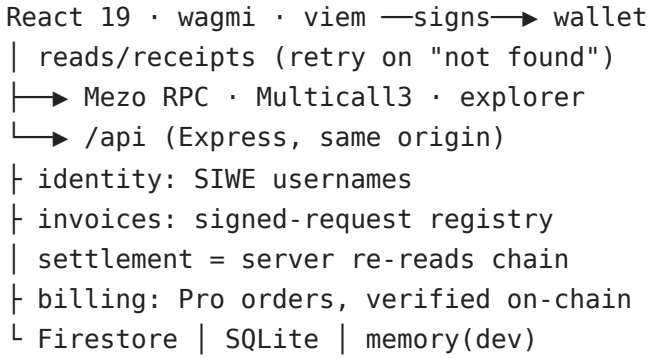
on-chain proofs verified (transfer, swap); signed settlement pending

Ready

Consent, onboarding, observation and case-study templates; `pilot:metrics` counts only verifiable transactions; `check:submission` stays BLOCKED until real participants exist

PILOT GOAL (NEXT 4 WEEKS)	TARGET	MEASURED BY
Consenting testnet merchants	3–5	consent forms
Verified signed-invoice settlements · distinct payers	≥ 30 · ≥ 15	`npm run pilot:metrics` (chain-derived only)
Interviews · usability sessions · willing to pay ≥ 9 MUSD	10 · 5 · ≥ 3	docs/WAVE2_PRODUCT_VALIDATION.md

Non-custodial, composable, verified at every step



No custom smart contracts. No keys, no custody.

Controls

- Chain and account re-checked before every prompt; double-submit locks
- Exact approvals, re-quote + simulate, router minimum, 30 s quotes
- Status is never client-set; incomplete search = unknown, not unpaid
- Invoices keyed (recipient, id); SIWE session for listing
- CSP, rate limits, deny-by-default Firestore rules

Tests (2026-11-08)

305 unit/integration/API · 26 Firestore emulator · 26 browser e2e (production build, live testnet reads, never broadcasts) · 7 live read-only · ESLint 0 warnings · CI on main, every PR and daily. Fixed this wave: a pay-while-pending double-payment bug. **Not audited.**

DIFFERENTIATION

Why Cambio, and why on Mezo

	WALLET SEND + PASTED ADDRESS	HOSTED CRYPTO CHECKOUT	CAMBIO
Tamper-evident request	No	Hosted page	**Merchant's EIP-712 signature, checked in the payer's browser**
Proof of payment	Screenshot	Provider dashboard	**Calldata-bound ID; anyone can verify**
Pay a dollar invoice holding only BTC	Manual DEX trip	Varies	**Fund & Pay on Mezo's MUSD/BTC pool**
Custody	None	Often custodial	**Non-custodial**
Mezo context	None	MUSD/Mezo support not verified by us	**PriceFeed, trove health, borrowing preview**

Comparison based on public product categories, not exhaustive testing. Cambio extends Mezo's own app into payments between people; it does not compete with borrowing or yield.

ROADMAP · SOLO BUILDER

Next: real merchants, real proofs, then mainnet

By Dec 15

Signed-settlement proof · redeploy · 3–5 merchant pilot with chain-derived metrics · 10 interviews

By Feb 15

Enable Merchant Pro in production · reminders and webhooks · Telegram bot that creates signed invoices

By May 15

Audited checkout contract (single-use invoices, fees) · optional borrowing route · mainnet only after review

Lushegs, solo developer: product, design and engineering. github.com/Lushegs01

Live: cambio-five.vercel.app · Repo: github.com/Lushegs01/cambio · Track 2: Access & Distribution · Mezo Testnet